



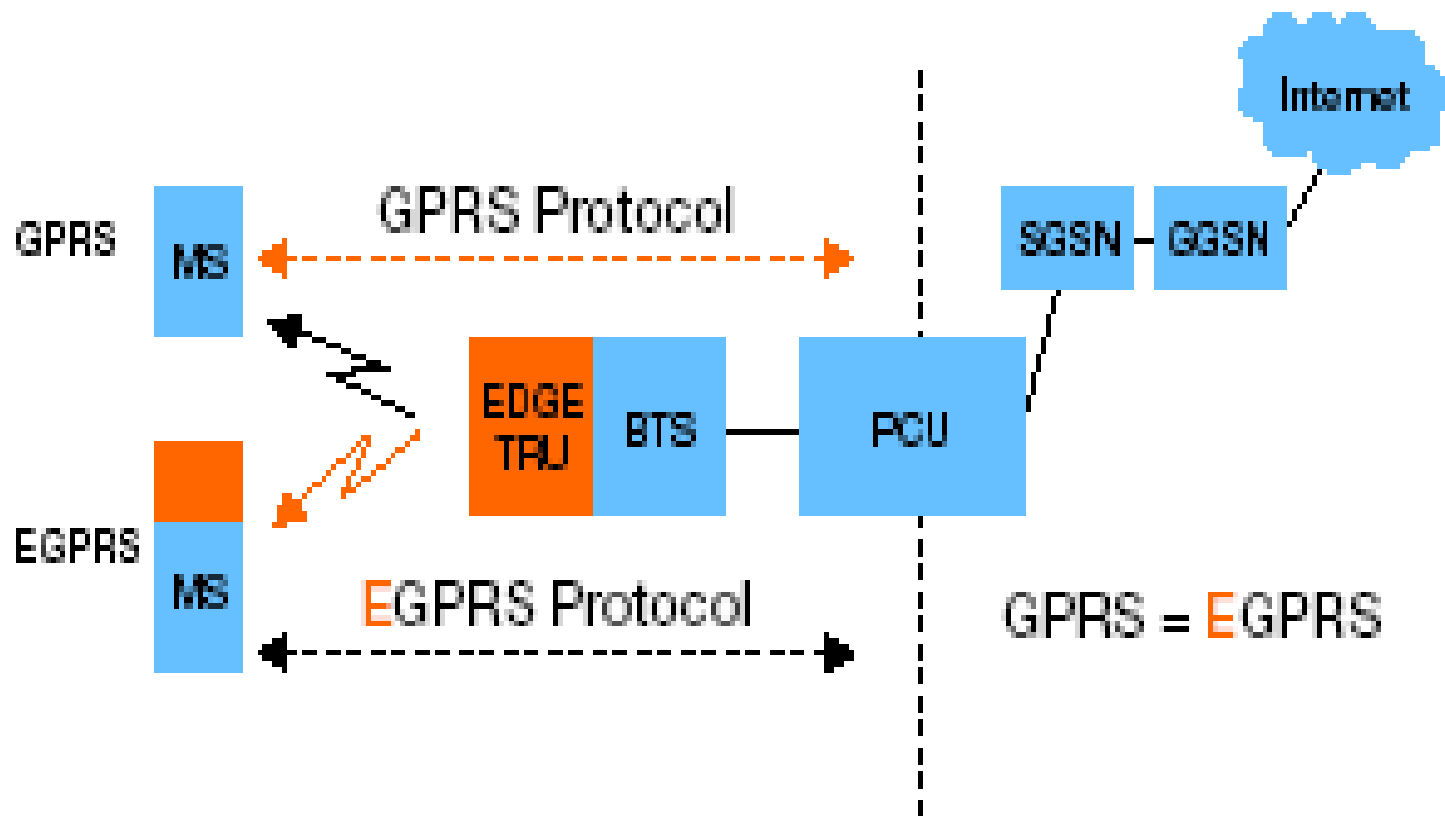
EDGE Technology

Introduction

- EDGE - Enhanced Data rates for Global Evolution
- Objective:
 - to increase data transmission rates and spectral efficiency
 - to facilitate new applications and increased capacity for mobile use

EDGE better than GPRS , how?

- GPRS - allows 115 kbps
theoretically up to 160 kbps
- EDGE - allows 384 kbps
theoretically up to 473.6 kbps
- new modulation technique
- error tolerant tx methods
- improved link adaptation mechanisms



LEGEND

BTS Base Station

EGPRS Enhanced GPRS

GGSN Gateway GPRS support node

MS Mobile station

PCU Packet control unit

SGSN Serving GPRS support node

TRU Transceiver unit

Base Station Subsystems

- Composed of Base Station Transceiver (BTS) and Base station controller (BSC)
- BTS - radio transmission and reception equipment. It provides coverage to a particular geographic area and is controlled by the BSC
- BSC -- handles the medium access and radio resource scheduling

Serving GPRS Support

(SGSN)

- Controls GPRS service in a particular geographical coverage area
 - provides a point of attachment for the GPRS mobiles. A logical link is established between the mobile station and the SGSN, via the base station
 - responsible for the transport and delivery of packets to and from the user
 - is responsible for validating the mobile stations, before they are allowed access to the GPRS system, and also performing security functions such as authentication and ciphering

Gateway GPRS Support Node (GGSN)

- provides connectivity to the external packet data networks
- primary role is to route data to the mobile stations at their current points of attachment
- Once the mobile station activates its packet data address, the mobile station is registered with the corresponding GGSN. The GGSN maintains a routing table associating the active GPRS mobiles in the system with a particular SGSN.

EDGE vs GPRS – a comparison of technical data

	GPRS	EDGE
Modulation	GMSK	8-PSK/GMSK
Symbol rate	270 ksym/s	270 ksym/s
Modulation bit rate	270 kb/s	810 kb/s
Radio data rate per time slot	22,8 kb/s	69,2 kb/s
User data rate per time slot	20 kb/s (CS4)	69,2 kb/s (MCS9)
User data rate (8 time slots)	160 kb/s	473,6 kb/s
	(182,4 kb/s)	(553,6 kb/s)

Channel

Coding

- Both the networks support adaptive channel coding—
- for good channel conditions maximum throughput is aimed at
- for poorer channel conditions error control bits are used to reduce errors and hence reduce the number of re transmissions

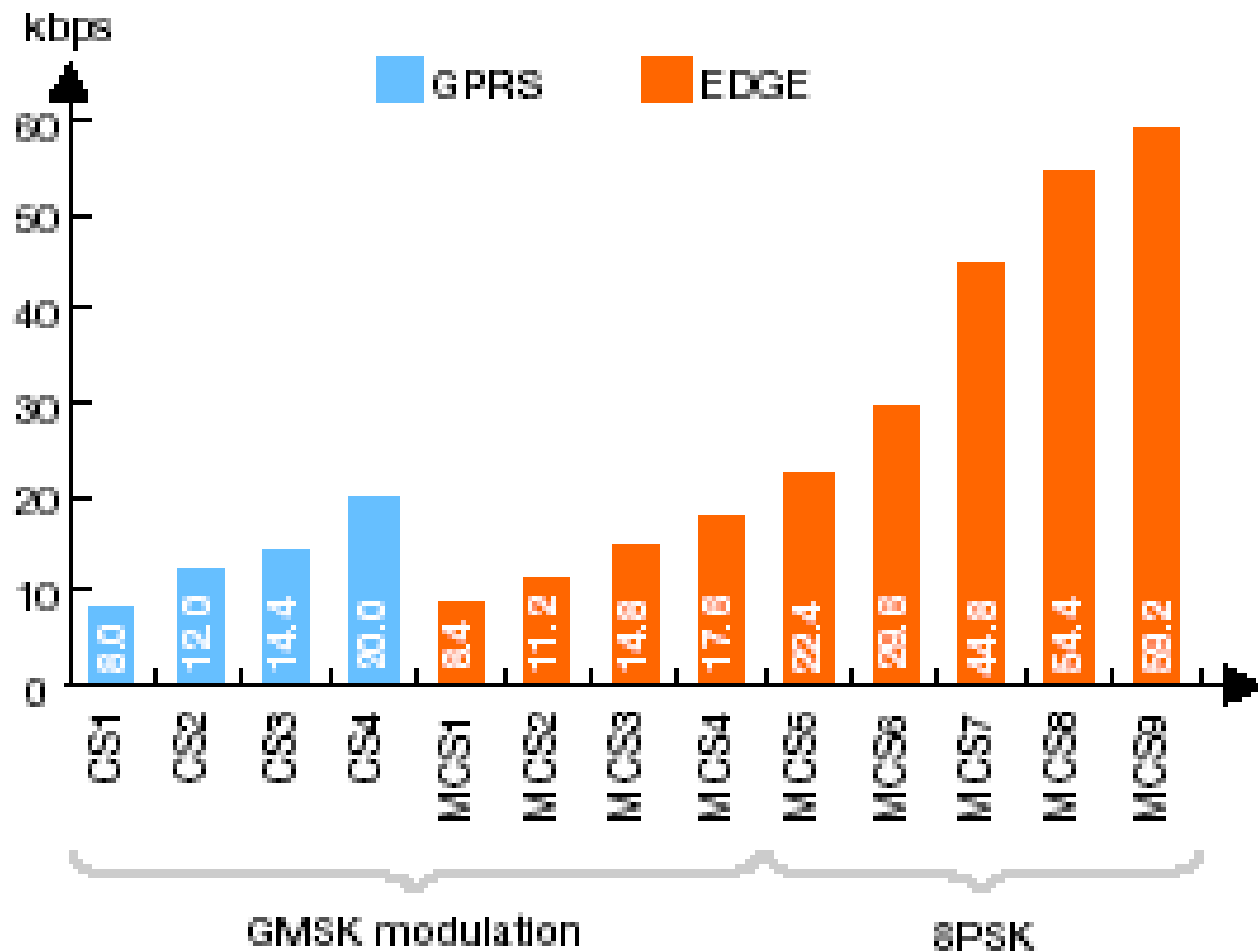
C o d i n g

S c h e m e s

- GPRS - 4 coding schemes - CS1 to CS4

Each has different amounts of error correcting coding that is optimized for different radio environments.

- EDGE - 9 coding schemes - MCS1 to MCS9
 - MCS1 to MCS4 - use GMSK
 - MCS5 to MCS9 - use 8PSK



Interleaving

- ❑ Blocks are broken up and interleaved between bursts. To receive the whole data all the bursts need to be received.
- ❑ For GPRS – 1 in 4 bursts.
- ❑ For EDGE – 1 in 2 bursts.

Incremental Redundancy

- Initially uses MCS9 and if noise level increases then coding scheme is changed (adaptive coding) to sustain the noise level
- If Information received incorrectly then Retransmission takes place and combined in the receiver with the previously received information.
- Procedure is repeated until information is successfully decoded.

Modulation

technique

- Modulation technique in GPRS - Gaussian Minimum Shift Keying

Modulation

technique

- ❑ Modulation standard for EDGE - 8-Phase shift Keying (8PSK).
- ❑ 8PSK - linear modulation technique - three consecutive bits are mapped onto one symbol.
- ❑ The symbol rate remains the same but each symbol represents 3 bits instead of one.
- ❑ Total data rate increases by a factor of three.

Modulation Techniques

- Distance between different symbols is shorter using 8PSK.
- Increases misinterpretation.
- Under poor radio conditions extra bits will be added for error correcting.
- Only under very poor conditions GMSK is more efficient.

RLC

- Error correction is done by selective retransmission of only the erroneous frames
- The correctly received frames are buffered until the erroneous frames are correctly received

P a c k e t

Handling

- EDGE - Has the ability to retransmit an erroneous packet with a more robust coding scheme
- GPRS -- Once packets have been sent, they must be retransmitted using the original coding scheme even if the radio environment has changed.

EDGE benefits

- Short-term benefits -
 - EDGE triples the capacity of GPRS.
 - Boosts performance of existing application and enables services such as multimedia.....
 - Enables transceiver to carry more voice/data traffic.

EDGE benefits

- With EDGE, operators can offer more wireless data applications, including wireless multimedia, e-mail, web infotainment and positioning services, for both consumer and business users.
- Subscribers will be able to browse the Internet on their mobile phones, personal digital assistants or laptops at the speed as on stationary personal computers